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A new model of soft tissue with constraints for interactive surgical simulation

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ABSTRACT

Background and objectives: An accurate and real-time model of soft tissue is critical for surgical simulation for which a user interacts haptically and visually with simulated patients. This paper focuses on the real-time deformation model of brain tissue for the interactive surgical simulation, such as neurosurgical simulation.

Methods: A new Finite Element Method (FEM) based model with constraints is proposed for the brain tissue in neurosurgical simulation. A new energy function of constraints characterizing the interaction between the virtual instrument and the soft tissue is incorporated into the optimization problem derived from the implicit integration scheme. Distance and permanent deformation constraints are introduced to describe the interaction in the convexity meningioma dissection and hemostasis. The proposed model is particularly suitable for GPU-based computing, making it possible to achieve real-time performance.

Results and conclusions: Simulation results show that the simulated soft tissue exhibits the behaviors of adhesion and permanent deformation under the constraints. Experiments show that the proposed model is able to converge to the exact solution of the implicit Euler method after 96 iterations. The proposed model was implemented in the development of a neurosurgical simulator, in which surgical procedures such as dissection of convexity meningioma and hemostasis were simulated.

1. Introduction

Neurosurgery is one of the most challenging, complex and technically demanding surgical procedures. A long period of training practices are required to ensure that neurosurgical residents gain proficiency in microsurgical skills, such as ventriculostomy, tumor debulking, hemostasis and microdissection, which are vital for avoiding intraoperative risks [1–3]. Studies have demonstrated that surgical simulation systems play important roles in the acquisition of surgical skills, which enables young neurosurgical beginners to gain hands-on experiences through practising on risk-free environments [4–7]. An interactive neurosurgical simulation system allows neurosurgical students to manipulate soft tissue models through virtual instruments, which produces complex physical phenomena such as elastic or plastic deformation, contact and adhesion [8–11].

One way to achieve accurately and real-time modeling of soft tissues is incorporating the FEM-based model using an optimization implicit Euler method [12,13]. A descent method can be used to solve the deformation model, which is suitable for GPU-based computing. Both anisotropic and viscoelastic properties of brain tissues can be incorporated into the deformation model. However, these models can only describe the dynamics of free deformable bodies subject to external gravity with optionally static nodes [14]. Interactions between brain tissues and rigid neurosurgical instruments, which take place during neurosurgical procedures, cannot be properly simulated by these models.

The FEM model is one of the most commonly used physically based approaches to model soft tissues in the surgical simulation. Continuum constitutive equations are used to govern the elastic behaviors of the soft tissues, which can provide a high degree of mechanical realism [15]. Partial differential equations (PDEs) of the dynamic system derived from the FEM model are solved by integrating procedures. Several works have been proposed to incorporate the FEM deformation model into the development of a surgical simulator [16–19]. In [17], Delorme et al. developed a physics-based virtual microneurosurgery simulator, in which the FEM model with an explicit time integration scheme is used to model the deformation of brain tissue. In [19], an open-source finite element toolkit, NiftySim, is proposed for simulation of soft tissue biomechanics. Contacts between multiple biological tissues,

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self-collisions and contacts between biological tissues and surgical instruments can be simulated in this toolkit. In [18], an open-source C++ library for interactive medical simulation is proposed, in which the internal mechanical behavior of biological tissues can be computed using the FEM model. Interactions between virtual objects, such as contact, collision response were handled by constraint solvers. While the FEM model provides the accurate description of the soft tissue deformation, low computational efficiency is the bottleneck in developing a surgical simulator which has the requirement of real-time performance.

Position Based Dynamic (PBD) method is another commonly used approach to model soft tissues. Position changes are computed directly in each simulation time step. Overshooting problem of explicit integration schemes can be avoided in contrast with classical force based simulation approaches [20,21]. Continuous materials can be simulated by combining a continuum-based formulation with the position based method [21–23]. In [21], a set of constraints for the strain energy are introduced into the position based method to model elastic behaviors of continuous materials. Complex physical effects such as lateral contraction, anisotropy and elastoplasticity of deformable objects can be simulated. In [22], a set of constraints which forcing the entries of the Green - St. Venant strain tensor to zero are introduced into the PBD framework. Stretch and shear deformations of continuous materials can be controlled independent of the tessellation of the mesh. The anisotropic behavior can also be simulated by varying the stiffness values corresponding to the individual strain coefficients.

Controllability is the main advantage of the PBD method and various kinds of interactions such as collision, friction and volume conservation can be handled easily with constraints, which makes it well-suited for interactive simulations. The position based method have become very popular in computer graphics because of its simplicity, controllability and robustness. Several works have been proposed to describe the elastic behaviors of soft tissues with a position based dynamic framework in surgical simulations [24,25]. In [26], the position based method is used to model objects collision, friction and volume conservation of soft tissues in an interactive dynamic simulation environment. In [27], Sui et al. proposed a hybrid deformation model to simulate the mechanical behavior of biological tissues surrounding the rectum, which based on a multi-layer structure and a position based dynamic framework. Interactions between different objects can be easily handled through introducing new constraints [28]. However, these methods are generally not as accurate as physic-based methods but provide visual plausibility.

Optimization implicit Euler method is another successful way of solving the PDEs of the dynamic system derived from the force-based deformation model. The dynamic system can be converted into an unconstrained nonlinear optimization problem [12,13,29–33]. Various optimization strategies can be implemented, which can provided a high computational efficiency. In [29], Liu et al. expressed the implicit Euler integration method as an energy minimization problem and introduced an new energy potential function of the spring mass system. A block coordinate descent method is used to solve the minimization problem, which converges to the exact solution of the implicit Euler method. In [30], a projective implicit Euler solver was proposed to solve the optimization problem, which build a bridge between the Finite Element method and the PBD method. In [31], Gast et al. proposed to use a constrained minimization method to simulate the collision between virtual objects. In [12], Wang et al. proposed a new gradient descent method with Jacobi preconditioning and Chebyshev acceleration to solve the nonlinear optimization problem derived from the implicit time integration scheme. In [13], the optimization implicit Euler method was implemented into the development of a neurosurgical simulator, in which the mechanical properties of brain tissue such as anisotropy and viscoelasticity were incorporated into the deformation model. The descent method proposed in [12] is particularly suitable for GPU-based parallel computing, which make it possible to achieve real-time performance for computation-intensive neurosurgical simulation. While these methods are promising, they are still unable to simulate various interactions between multiple soft tissues and instruments properly.

In this paper, we introduce a new soft tissue deformation model with constraints for neurosurgical simulation. Specifically, a new energy function of constraint is introduced into the optimization problem derived from the implicit Euler integration scheme; distance and permanent deformation constraints are introduced to characterize the interactions in the convexity meningioma dissection and hemostasis. The descent method is used to solve the optimization problem, which provides high computational efficiency. Examples are presented to assess the performance of the developed deformation model. Simulation results show that the interactions such as adhesion and permanent deformation can be properly simulated by the presented deformation model. In the meantime, the GPU-based implementation of the developed deformation model improves computational efficiency significantly over CPU-based FEM models with the implicit integration scheme. The developed model was incorporated into the development of a neurosurgical simulator, in which neurosurgical procedures such as dissection of convexity meningioma and hemostasis were simulated. The simulator provides a relatively high degree of visual realism with a visual refreshment rate of 31.1 frames per second on a regular PC.

The novelty of this work exists in at least two aspects: (1) A new energy function of constraints is introduced into the optimization problem, which characterizes well the interactions between surgical instruments and soft tissues; (2) Both distance and permanent deformation constraint functions are integrated into the model of soft tissue, making it particularly useful to model the behaviors of connections between different types of soft tissues during the surgical process. The results are very important for the simulation of neurosurgical procedures, such as hemostasis and tumor dissection.

The reminder of this paper is organized as follows. In Section 2, the FEM-based model using the optimization implicit Euler method is briefly explained. In Section 3, we show how the new energy function of constraint is introduced into the deformation model. Implementation and simulation results are provided in Section 4. A brief discussion is presented in Section 5. Conclusion and future work are given in Section 6.

2. Background

In the FEM-based deformation model, the deformation of soft tissue can be formulated as a function $\phi : \mathbb{R}^3 \to \mathbb{R}^3$, which maps an undeformed configuration into a deformed configuration. The Linear Tetrahedron (LT) technique is often used to discrete the deformable object into a number of finite elements. When the soft tissue undergoes deformation, a strain energy function W is used to characterize the stress-strain relationship of the soft object. In each tetrahedron mesh $\mathcal{T}_i$, the elastic forces associated with each individual mesh vertex are the negative gradient of the strain energy function with respect to the corresponding degree of freedom, which can be written as

$$f_i(x_j) = -\frac{\partial W_i}{\partial \vec{x}_j} \tag{1}$$

where $f_i(x_j)$ is the forces applied on vertex j which is contributed by its immediate neighbor element $\mathcal{T}_i$.

For the FEM-based deformation model, implementation of the implicit time integration scheme results in a large system of nonlinear equations, which can be converted into a nonlinear optimization problem [12,13]. Consider the tetrahedron mesh consisting of N vertices with position vector $\boldsymbol{x} = (\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_N)^T$, and let $\boldsymbol{v} = (\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_N)^T$ be the velocity vector of the mesh vertices. The dynamic system derived from the FEM-based model evolves in time according to the Newton's laws of motion through a discrete set of time samples. The deformation of soft tissue from time step t to $t+1$ can be calculated through an implicit time integration scheme, as:

$$\begin{aligned} \boldsymbol{x}_{t+1} &= \boldsymbol{x}_t + h\boldsymbol{v}_{t+1} \\ \boldsymbol{v}_{t+1} &= \boldsymbol{v}_t + h\boldsymbol{M}^{-1}\boldsymbol{f}(\boldsymbol{x}_{t+1}) \end{aligned} \tag{2}$$

where h is the simulation time step size, $\boldsymbol{M} \in \mathbb{R}^{3N\times 3N}$ is the mass matrix of mesh vertices, $\boldsymbol{f} \in \mathbb{R}^{3N}$ is the elastic forces applied on each mesh vertices. By combing the two equations, we derive a single nonlinear system:

$$\boldsymbol{M}(\boldsymbol{x}_{t+1} - \boldsymbol{x}_t - h\boldsymbol{v}_t) = h^2\boldsymbol{f}(\boldsymbol{x}_{t+1}) \tag{3}$$

Since the elastic forces are the negative gradient of the strain energy as $\boldsymbol{f}(\boldsymbol{x}) = -\partial W(\boldsymbol{x})/\partial \boldsymbol{x}$, where $W(\boldsymbol{x})$ is the total potential energy evaluated at $\boldsymbol{x}$, the nonlinear dynamic system can be converted into an unconstrained nonlinear optimization problem: $\boldsymbol{x}_{t+1} = \arg\min \epsilon(\boldsymbol{x})$

$$\epsilon(\boldsymbol{x}) = \frac{1}{2h^2}(\boldsymbol{x} - \boldsymbol{s}_t)^T\boldsymbol{M}(\boldsymbol{x} - \boldsymbol{s}_t) + W(\boldsymbol{x}) \tag{4}$$

where $\boldsymbol{s}_t = \boldsymbol{x}_t + h\boldsymbol{v}_t$, which is related to the state of the discretized model at time t.

Several kinds of solvers have been proposed to solve the nonlinear optimization problem. In the PBD method, a Gauss–Seidel-like solver is used to minimize the object function $\epsilon(\boldsymbol{x})$. However, the inertial term $(\boldsymbol{x} - \boldsymbol{s}_t)^T\boldsymbol{M}(\boldsymbol{x} - \boldsymbol{s}_t)$ is not taken into account. A local/global solver is proposed to reduce specially designed energy potentials in the Projective Dynamic Method [30]. Another method to solve the optimization problem is the descent method [12,13], of which the key question is how to find an effective descent direction with low computational cost. Iterative procedures are performed in the descent method to minimize the object function $\epsilon(\boldsymbol{x})$.

The descent method used in our proposed deformation model is shown in Algorithm 1. At time step $t+1$, the vertex position

Algorithm 1 Deformation algorithm of proposed soft tissue deformation model.

Input: Geometric model of soft tissue
Output: Position of each vertex
1: Initialize $\boldsymbol{x}^{(0)}$
2: **for** $l = 0$ to $L-1$ **do**
3: Calculate the gradient and Hessian matrix of $\epsilon(\boldsymbol{x})$ at $\boldsymbol{x}^{(l)}$
4: Get descent direction: $\Delta\boldsymbol{x}^{(l)} = -diag^{-1}(\boldsymbol{H}^{(l)})\nabla\epsilon(\boldsymbol{x}^{(l)})$
5: Adjust the step length $\alpha^{(l)}$
6: Update vertex position: $\boldsymbol{x}^{(l+1)} \leftarrow \boldsymbol{x}^{(l)} + \alpha^{(l)}\Delta\boldsymbol{x}^{(l)}$
7: **end for**
8: Obtain the new position of each vertices

$\boldsymbol{x}_{t+1}$ is calculated by solving the optimization problem described in Eq. (4). The descent method contains four steps. *Step* 1, the initial vertex position $\boldsymbol{x}^{(0)}$ is predicted through an explicit integration scheme $\boldsymbol{x}^{(0)} = \boldsymbol{x}_t + h\boldsymbol{v}_t$, which help reduce the object function $\epsilon(\boldsymbol{x})$ to a low level, as shown in line (1). *Step* 2, the descent direction at iteration step l is calculated. The estimate $\boldsymbol{x}^{(l)}$ is already known. In line (3), the gradient and Hessian matrix of the object function $\epsilon(\boldsymbol{x})$ are evaluated at $\boldsymbol{x}^{(l)}$. In line (4), a new descent direction is defined as:

$$\Delta\boldsymbol{x}^{(l)} = -diag^{-1}\left(\boldsymbol{H}^{(l)}\right)\nabla\epsilon\left(\boldsymbol{x}^{(l)}\right) \tag{5}$$

where $\boldsymbol{H}^{(l)}$ is the Hessian matrix of $\epsilon(\boldsymbol{x}^{(l)})$. *Step* 3, given the descent direction $\Delta\boldsymbol{x}^{(l)}$, the step length $\alpha^{(l)}$ is gradually reduced until satisfied the first Wolfe condition [12]:

$$\epsilon\left(\boldsymbol{x}^{(l)} + \alpha^{(l)}\Delta\boldsymbol{x}^{(l)}\right) < \epsilon\left(\boldsymbol{x}^{(l)}\right) + c^{(l)}\alpha^{(l)}\Delta\boldsymbol{x}^{(l)} \cdot \nabla\epsilon\left(\boldsymbol{x}^{(l)}\right) \tag{6}$$

in which c is a control parameter. *Step* 4, the estimate of the vertex position $\boldsymbol{x}^{(l+1)}$ is updated, as shown in line (6). *Step* 2 - *Step* 4 are iteratively performed, which could help to reduce the objective function $\epsilon(\boldsymbol{x})$ to minim. Finally, the new vertex position $\boldsymbol{x}_{t+1}$ at time step $t+1$ can be obtained after 96 iterations.

The descent method achieves accurate and efficient description of the soft tissue deformation, which meets the basic requirements of the surgical simulation. However, there are lots of interactive situation in the interactive neurosurgical simulation, such as adhesion of multi soft tissues, cutting, suction and permanent deformation, which cannot be properly simulated.

3. A new brain tissue deformation model with constraints

So far we have described the dynamics of a free deformation body subject to external gravity with optionally static nodes. Soft tissues can deform under the manipulation such as traction in the virtual environment, which satisfies the basic requirements of surgical simulation. However, a neurosurgical simulator is supposed to help neurosurgeons practice various kinds of surgical procedures which contains different kinds of interactions between soft tissues and instruments. For example, dissection of convexity meningiomas is one of the classic situation for neurosurgical training exercise, where the meningiomas adheres to the surface of brain tissue. Hemostasis is another essential technique in the neurosurgery that demands neurosurgeons to use bipolar forceps to control blood loss, which would cause permanent deformation of soft tissues. Neurosurgical simulation is performed in a highly interactive environment where the free body equations of motion can be transiently changed [14]. Interactions and the phenomenon they produced should be properly simulated, which help to improve training outcomes by providing realistic visual feedback.

In FEM-based method and PBD method, interactions between virtual objects are usually treated as constraints [28]. The position based method can be interpreted as descent method under a nonlinear optimization framework derived from implicit integration scheme [30]. Inspired by that, the constraint function in PBD method is adopted into the optimization implicit method. In this paper, a constraint function, which characterizes the interactions between virtual objects, is introduced into the optimization implicit method. The proposed deformation model can provide a high realistic description of soft tissue deformation in an interactive environment.

When there are interactions between virtual objects, they can be formulated as a set of constraints $\boldsymbol{C}(\boldsymbol{x}) = [C_1(\boldsymbol{x}), C_2(\boldsymbol{x}), \ldots, C_m(\boldsymbol{x})]^T$. The key to our proposed deformation method is to introduce a new extra constraint term into the optimization problem. In order to incorporate the constraints into the optimization problem, we define a new energy potential $U(\boldsymbol{x})$ as function of constraints. Inspired by Macklin et al. [34], the new energy potential can be formulated as

$$U(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{C}(\boldsymbol{x})^T\boldsymbol{\gamma}\boldsymbol{C}(\boldsymbol{x}) \tag{7}$$

where $\boldsymbol{\gamma}$ is a block diagonal matrix. The entries of matrix $\boldsymbol{\gamma}$ represent the stiffness of each constraint in the dynamic system derived from the FEM-based model. Given the potential energy function of interaction constraints, the equivalent constraint forces, which press the deformation model such that it is displaced and reaches a new rest state, can be written as the negative gradient of $U(\boldsymbol{x})$ with respect to $\boldsymbol{x}$,

$$\boldsymbol{f}_C = -\nabla_x U^T = -\nabla\boldsymbol{C}^T\boldsymbol{\gamma}\boldsymbol{C} \tag{8}$$

When new constraints are applied on the deformation model, the equivalent constraints forces are added into the nonlinear dynamic system described in (3), as

$$\boldsymbol{M}(\boldsymbol{x}_{t+1} - \boldsymbol{x}_t - h\boldsymbol{v}_t) = h^2 \boldsymbol{f}(\boldsymbol{x}_{t+1}) + h^2 \boldsymbol{f}_C(\boldsymbol{x}_{t+1}) \tag{9}$$

The nonlinear dynamic system with extra constraint forces can also be converted into a new nonlinear optimization problem: $\boldsymbol{x}_{t+1} = \arg\min \epsilon(\boldsymbol{x})$

$$\epsilon_C(\boldsymbol{x}) = \frac{1}{2h^2}(\boldsymbol{x} - \boldsymbol{s}_t)^T \boldsymbol{M}(\boldsymbol{x} - \boldsymbol{s}_t) + W(\boldsymbol{x}) + U(\boldsymbol{x}) \tag{10}$$

where $\boldsymbol{s}_t = \boldsymbol{x}_t + h\boldsymbol{v}_t$, $U(\boldsymbol{x})$ is the energy potential of constraints. In the simulation of soft tissue, nonlinear Eq. (9) are used to govern the dynamics of the deformable object. Elastic forces which applied on each vertices make the deformation model maintain dynamic equilibrium. As the constraint is introduced into the deformation model, the constraints forces will compel the deformable objects rest into a desired state.

In our proposed deformation model, the descent method is used to solve the optimization problem as described in Algorithm 1. At the beginning of each time step $t+1$, an initial estimate of $\boldsymbol{x}^{(0)}$ is predicted through performing an explicit time integration step. In each iteration step l, the descent direction at $\boldsymbol{x}^{(l)}$ is calculated as

$$\Delta\boldsymbol{x}^{(l)} = -diag^{-1}\left(\boldsymbol{H}_C^{(l)}\right)\nabla\epsilon_C\left(\boldsymbol{x}^{(l)}\right) \tag{11}$$

where $\boldsymbol{H}_C^{(l)}$ is the Hessian matrix of $\epsilon_C(\boldsymbol{x}^{(l)})$. A suitable step length $\alpha^{(l)}$ is adopted which satisfies the simplified first Wolfe condition, as

$$\epsilon_C\left(\boldsymbol{x}^{(l)} + \alpha^{(l)}\Delta\boldsymbol{x}^{(l)}\right) < \epsilon_C\left(\boldsymbol{x}^{(l)}\right) \tag{12}$$

in which the control parameter c is set to zero. Finally, the vertices move to the new estimated position along the descent direction $\boldsymbol{x}^{(l+1)} = \boldsymbol{x}^{(l)} + \alpha^{(l)}\Delta\boldsymbol{x}^{(l)}$. After L times of iterations, the final solution of the optimization problem is obtained, which is the new vertex position $\boldsymbol{x}_{t+1}$ at time step $t+1$.

The energy function of constraint, which characterizes the interactions between virtual objects in the interactive neurosurgical simulation, is incorporated into the optimization implicit method. For different kinds of interactions in the neurosurgical simulation, the constraints can be incorporated in the same way. In this paper, energy functions of constraint, including distance and permanent deformation constraints, are introduced.

3.1. Distance constraint

Dissection of convexity meningioma is one of the classical situations for neurosurgical training exercise, in which the brain tumor adheres to the surface of brain tissue. When neurosurgeon performs microdissection, the tumor is retracted with forceps to expose the tumor–brain interface. Neurosurgical instruments, such as microscissors, are often used to separate the brain tissue and the tumor. Simulation of convexity meningioma dissection could help young neurosurgeons to gain proficiency in microsurgical skills, which requires proper simulation of the adhesions between the brain and the meningioma as well as the separation process.

In this paper, we proposed a distance constraint to characterize the adhesion between the brain tissue and the tumor. When the tumor adheres to the brain, the distance between surfaces of brain and tumor keeps constant. In the deformation model, the initial distance between each pair of two attached vertices $\vec{x}_{q1}$ and $\vec{x}_{q2}$ is d_q. The distance constraint can be defined as

$$C_q(\boldsymbol{x}) = ||\vec{x}_{q1} - \vec{x}_{q2}|| - d_q \tag{13}$$

Given the distance constraint of each pair of attached vertices in the deformation model, we formulate the energy function of constraint as

$$\begin{aligned} U(\boldsymbol{x}) &= \frac{1}{2}\boldsymbol{C}(\boldsymbol{x})^T \boldsymbol{\gamma} \boldsymbol{C}(\boldsymbol{x}) \\ &= \frac{1}{2}\sum_{q=1}^{m} k_q\left(||\vec{x}_{q1} - \vec{x}_{q2}|| - d_q\right)^2 \end{aligned} \tag{14}$$

where $\boldsymbol{\gamma} = diag(k_1, k_2, \ldots, k_m)$. In the energy function of distance constraint, k_q represents the stiffness of each distance constraint.

The energy function of distance constraint is incorporated into the optimization problem (10). When solving the optimization problem in each time step, the descent direction in each iteration can be calculated as

$$\begin{aligned} \Delta\boldsymbol{x}^{(l)} &= -diag^{-1}\boldsymbol{H}_C^{(l)}\nabla\epsilon_C\left(\boldsymbol{x}^{(l)}\right) \\ &= -diag^{-1}\left(\boldsymbol{H}^{(l)} + \nabla^2 U\left(\boldsymbol{x}^{(l)}\right)\right)\left(\nabla\epsilon\left(\boldsymbol{x}^{(l)}\right) + \nabla U\left(\boldsymbol{x}^{(l)}\right)\right) \end{aligned} \tag{15}$$

where $\nabla\epsilon(\boldsymbol{x}^{(l)})$ is the gradient of total energy potential $\epsilon(\boldsymbol{x})$ without considering constraints evaluated at $\boldsymbol{x}^{(l)}$ and $\boldsymbol{H}^{(l)}$ is its Hessian matrix. The gradient of distance energy potential in (15) is calculated as

$$\nabla U(\boldsymbol{x}) = \nabla \boldsymbol{C}^T \boldsymbol{\gamma} \boldsymbol{C} \tag{16}$$

in which

$$\begin{aligned} \frac{\partial C_q(\boldsymbol{x})}{\partial \vec{x}_{q1}} &= \frac{\vec{x}_{q1} - \vec{x}_{q2}}{||\vec{x}_{q1} - \vec{x}_{q2}||} \\ \frac{\partial C_q(\boldsymbol{x})}{\partial \vec{x}_{q2}} &= \frac{\vec{x}_{q2} - \vec{x}_{q1}}{||\vec{x}_{q1} - \vec{x}_{q2}||} \end{aligned} \tag{17}$$

In (15), $\nabla^2 U(\boldsymbol{x})$ is the Hessian matrix of energy function of distance constraint, which can be written as

$$\nabla^2 U(\boldsymbol{x}) = \nabla^2 \boldsymbol{C}^T \boldsymbol{\gamma} \boldsymbol{C} + \nabla \boldsymbol{C}^T \boldsymbol{\gamma} \nabla \boldsymbol{C} \tag{18}$$

where

$$\begin{aligned} \frac{\partial^2 C_q(\boldsymbol{x})}{\partial \vec{x}_{q1}\partial \vec{x}_{q1}} &= \frac{\boldsymbol{I}}{||\vec{x}_{q1} - \vec{x}_{q2}||} - \frac{\left(\vec{x}_{q1} - \vec{x}_{q2}\right)\left(\vec{x}_{q1} - \vec{x}_{q2}\right)^T}{||\vec{x}_{q1} - \vec{x}_{q2}||^3} \\ \frac{\partial^2 C_q(\boldsymbol{x})}{\partial \vec{x}_{q2}\partial \vec{x}_{q2}} &= \frac{\boldsymbol{I}}{||\vec{x}_{q1} - \vec{x}_{q2}||} + \frac{\left(\vec{x}_{q1} - \vec{x}_{q2}\right)\left(\vec{x}_{q1} - \vec{x}_{q2}\right)^T}{||\vec{x}_{q1} - \vec{x}_{q2}||^3} \end{aligned}$$

When deformable objects adhere with each other, a set of attached vertex pairs can be identified. Distance constraints are imposed in order to keep the distance between each pair of vertices constant. When the deformable objects is separated with each other by the neurosurgical instruments, the constraints imposed on the vertex pair are removed respectively.

3.2. Permanent deformation constraint

Hemostasis is one the most important technical skills that must be mastered by neurosurgeons. Bipolar forceps are usually used to cauterize blood vessels to complete coagulation. The heat generated by bipolar forceps results to permanent deformation of brain tissue. Simulation of permanent deformation with the optimization implicit method can help to provide more realistic visual feedback in virtual neurosurgical training.

In this paper, we proposed a new permanent deformation constraint function to characterize the permanent deformation of brain tissue during the hemostasis procedure. When the brain tissue deforms from its reference configuration, the strain energy of deformed object can be written as $W(q) = \Psi(\boldsymbol{E}(q))$. From geometric point of view, $\Psi(\boldsymbol{E}(q)) = 0$ represents the undeformed configuration of the deformation model, of which the strain tensor $\boldsymbol{E}(q)$

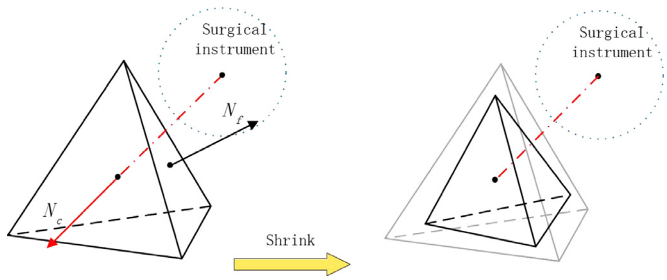


Fig. 1. Permanent deformation of the tetrahedron mesh. N_c is the contact normal. N_f is the surface normal of tetrahedron mesh.

equals to zero. The Green - St. Venant strain tensor of the deformed object is defined as

$$\boldsymbol{E} = \frac{1}{2}\left(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I}\right) \tag{19}$$

The diagonal entries of $\boldsymbol{E}_{ij}$ represent stretch and the off-diagonal entries $\boldsymbol{E}_{ij} = \boldsymbol{E}_{ji}$ shear both with respect the main axes in the material frame [22]. The stretch constraints are incorporated into the deformation model

$$C_p(x_1, x_2, x_3, x_4) = \boldsymbol{S}_{ii} - s_i \tag{20}$$

in which $\boldsymbol{S} = \boldsymbol{F}^T\boldsymbol{F}$ and s_i is the shrink ratio in each direction. From a geometric point of view, these three constraints would force the virtual object deform to the shrunk shape that the Green - St. Venant strain tensor goes back to the desired state.

The energy potential of permanent deformation constraint can be written as

$$U(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{C}(\boldsymbol{x})^T\boldsymbol{\gamma}\boldsymbol{C}(\boldsymbol{x}) \tag{21}$$

in which $\boldsymbol{\gamma}$ can be set as $\boldsymbol{I}$. When solving the optimization problem (10), the descent direction in each iteration can be calculated as

$$\begin{aligned}\Delta\boldsymbol{x}^{(l)} &= -diag^{-1}\boldsymbol{H}^{(l)}\nabla\epsilon_C\left(\boldsymbol{x}^{(l)}\right)\\ &= -diag^{-1}\boldsymbol{H}^{(l)}\left(\nabla\epsilon\left(\boldsymbol{x}^{(l)}\right) + \nabla U\left(\boldsymbol{x}^{(l)}\right)\right)\end{aligned} \tag{22}$$

where $\nabla\epsilon(\boldsymbol{x}^{(l)})$ is the gradient of total energy potential $\epsilon(\boldsymbol{x})$ without considering constraints evaluated at $\boldsymbol{x}^{(l)}$ and $\boldsymbol{H}^{(l)}$ is its Hessian matrix. The gradient of energy function of permanent deformation constraint is calculated as

$$\nabla U(\boldsymbol{x}) = \nabla\boldsymbol{C}^T\boldsymbol{\gamma}\boldsymbol{C} \tag{23}$$

in which

$$\nabla C_{ii} = \nabla(\boldsymbol{S}_{ii} - s_i) = 2\boldsymbol{f}_i\boldsymbol{c}_i^T \tag{24}$$

where $\boldsymbol{F} = [\boldsymbol{f}_1, \boldsymbol{f}_2, \boldsymbol{f}_3]$ is the deformation gradient, and $\boldsymbol{D}_m^{-1} = [\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3]$ is the reference shape matrix.

When the surgical instruments contact with brain tissue which cause permanent deformation, the tissues surrounding the contact point should shrink perpendicular to the contact surface. As shown in Fig. 1, the contact normal N_c passes through the contact point of surgical instrument and the center of the tetrahedron mesh in brain tissue model. In order to calculate the shrink ratio in the world coordinate, a new local coordinate system is established. The direction of N_c is set as one axis of the local coordinate system. The other two axis (N_a, N_b) are generated through the face normal (N_f) of the brain tissue, as $N_a = N_f \times N_c$ and $N_b = N_c \times N_a$. The shrink ratio s_l in world coordinate system can be calculated as

$$\begin{bmatrix} s_x & 0 & 0\\ 0 & s_y & 0\\ 0 & 0 & s_z\end{bmatrix} = \boldsymbol{R}^{-1}\begin{bmatrix} \sigma_x & 0 & 0\\ 0 & \sigma_y & 0\\ 0 & 0 & \sigma_z\end{bmatrix}\boldsymbol{R} \tag{25}$$

where σ_x, σ_y and σ_z are the shrink ratio in local coordinate system and $\boldsymbol{R}$ is the rotation matrix which transforms the initial normal strain direction to the local coordinate system of the contact point.

4. Simulation results

In order to demonstrate the effectiveness and validity of the proposed deformation model, tests on three-dimensional geometric models are performed. Various kinds of hyperelastic material models [35] that describe the elastic behavior of brain tissue can be implemented in our proposed deformation model. In our tests, the incompressible Mooney–Rivlin model is used to predict the deformation of brain tissue. Constraints of distance and permanent deformation are incorporated into the examples respectively. The proposed model is implemented with GPU using descent method as describe in Algorithm 1. The deformation model is incorporated into the development of a neurosurgical simulator.

4.1. Simulation of distance constraint

Two examples of the proposed deformation model with distance constraint are presented. In the first example, the cube model contains 882 vertices and 2880 tetrahedrons. The top of the cube model is attached on a fixed plane. Distance constraints are imposed on the attached vertices of the two cubes. In the second example, a liver model, which contains 2578 vertices and 8837 tetrahedrons, is presented. A gallbladder is attached on the surface of the liver. External manipulations are imposed on the models, as shown in Figs. 2 and 3.

In the first example, the initial state of the cube model is shown in Fig. 2(a). Distance constraints are imposed, as indicated with white lines. The initial distance d_i between each pair of attached vertices is 2 mm. External displacement are imposed on the bottom edge of the cube model, as shown in Fig. 2(b) and c. When the external traction is released, the cube model goes back to its rest state under the effect of gravity, as shown in Fig. 2(d). The second example is shown in Fig. 3. When the liver model is manipulated by a virtual instrument, the gallbladder keeps attached on the surface of the liver under the effect of distance constraints.

Simulation results show that the distance constraints keep the vertex pairs remain attached when external manipulation is performed on the virtual objects. It provides a way to describe the attachment of soft tissues in the surgical simulation. The distance constraint has been incorporated into the development of a neurosurgical simulator.

4.2. Simulation of permanent deformation constraint

An example of a cube model of edge length 0.05 m with permanent deformation constraint is presented. The cubic deformation model contains 10,648 vertex and 46,305 tetrahedrons. Vertices in the bottom of the cube model are fixed. An influence area parameter ξ is used to determine which tetrahedron mesh the permanent deformation constraint is imposed on. When a virtual surgical instrument contacts with the cube model, permanent deformation constraints are imposed on the tetrahedron meshes of which the distance from the contact point is closer than ξ. In this example, the influence area parameter ξ is set to 7.5 mm. Process of the permanent deformation as well as the elastic deformation of the cube model are shown in Fig. 4.

The initial state of undeformed cube model is shown in Fig. 4(a). When the hemostasis procedure is performed in neurosurgery, the bipolar forceps is used to cauterize the brain tissue, which cause permanent deformation of the brain tissue. As shown in Fig. 4(b) and (c), the virtual surgical instrument contacts

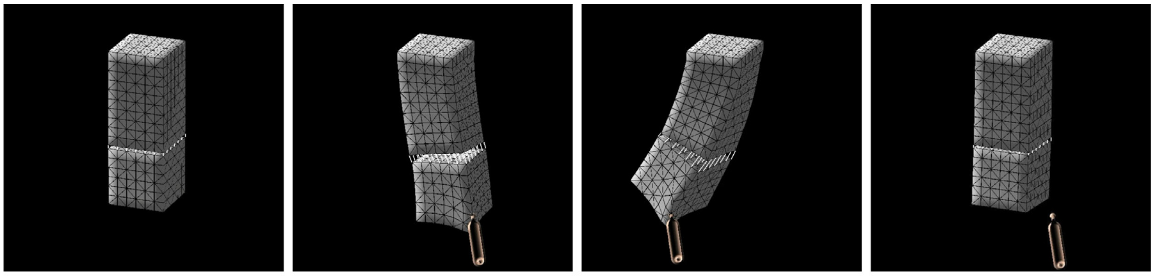


Fig. 2. Deformation of cube models with distance constraint: (a) shows the undeformed models, (b)–(d) show the deformation of cube models under external manipulation.

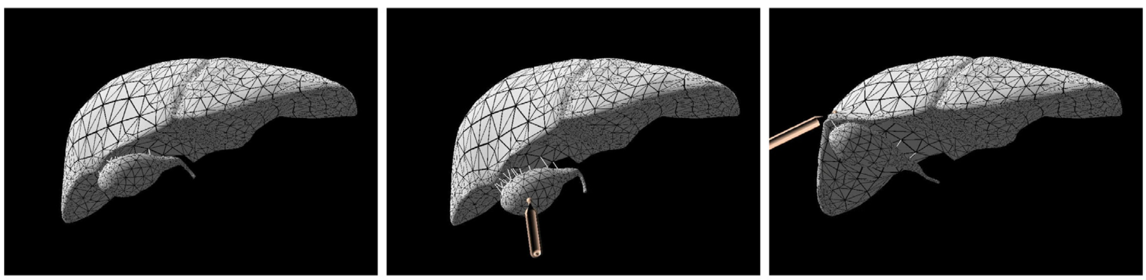


Fig. 3. Deformation of a liver model with distance constraint: (a) shows the undeformed liver model, (b) and (c) show the deformation of liver model under external manipulation.

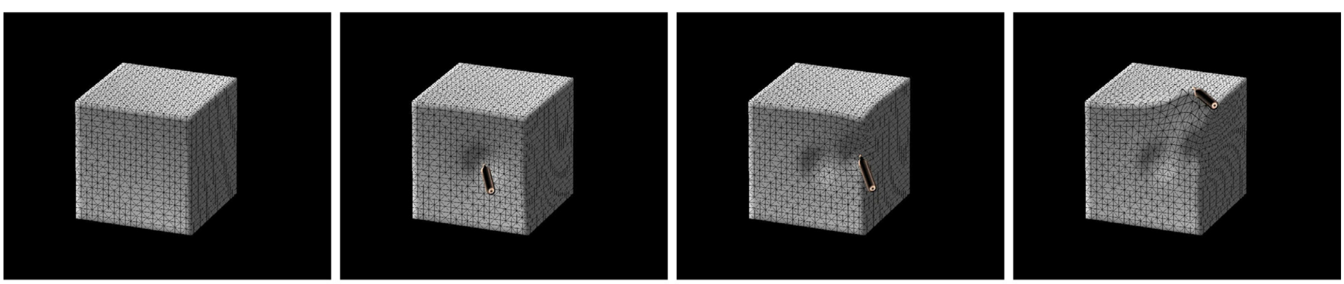


Fig. 4. Deformation of a cube model with permanent deformation constraint: (a) shows the undeformed brain tissue, (b)–(d) show the deformation of brain tissue under the effect of permanent constraint.

with the cube model. The contact point is selected as the mesh vertex of the virtual surgical instrument, which interacts with the surface of the cubic model. Tetrahedron meshes in the influence area are identified by the distance criterion. The permanent deformation constraints are imposed into the tetrahedrons in the influence area. The equivalent permanent deformation constraint forces presses the cube model such that it is displaced and reaches a new rest state. The shrink ratio of each tetrahedron mesh is calculated as described in Section 3.2. The contact normal (N_c) can be calculated through the contact point and the center of the tetrahedron mesh. The face normal N_f of the cube model is calculated through the coordinate of the triangle mesh in each time step. Then the local coordinate system of contact point is created and the shrink ratio s_l is calculated respectively. The shrink ratio σ in local coordinate system is 0.2. As shown in Fig. 4(d), external traction is imposed on the cube model and the distortion caused by cauterization keeps permanent.

4.3. Computational performance assessment

In order to assess the performance of our proposed deformation model, the computation time of a single time step is measured. As described in Algorithm 1, the descent method with GPU-base implementation is used to solve the optimization problem in our proposed deformation model. In each time step, the vertex position $\boldsymbol{x}^{(0)}$ is initialed with an explicit time integration. The descent direction is calculated and the vertex position is updated along the descent direction in every iteration as descried in Section 3. All of the tests use $t = 1/30$s as the time step and run 96 iterations per time step. The tests ran on an personal computer with an Intel(R) Core E5-2650 v4 CPU, 32GB RAM and a NVIDIA GeForce GTX TITAN Xp graphics card with 3840 cores.

The solution time of our proposed deformation is provide in Table 1. Simulation results show that our proposed deformation model with constraints achieves high computational efficiency. Compared with FEM-based deformation model using implicit integration scheme, our model is compatible to GPU-based computation, makes it suitable for computation-intensive neurosurgical simulation. The energy function of constraint incorporated into our proposed deformation model costs few computational resources in each time step. After the constraints are removed, they don't involve in the computation any more. In general, a visual refreshment rate of 30 Hz can meet the requirement of real-time performance in virtual surgical simulation. A refreshment of 31.1 frames per second can be achieved in our developed neurosurgical simulator.

Table 1
Solution time of the deformation model.

Method	Number of tetrahedron	Constraints number	Execution time (ms)
Our model	2880	0	20.47
	13,169	0	31.46
Our model	2880	49	20.82
	13,169	475	32.12
CPU-based model	2880	0	48.60
	13,169	0	249.74

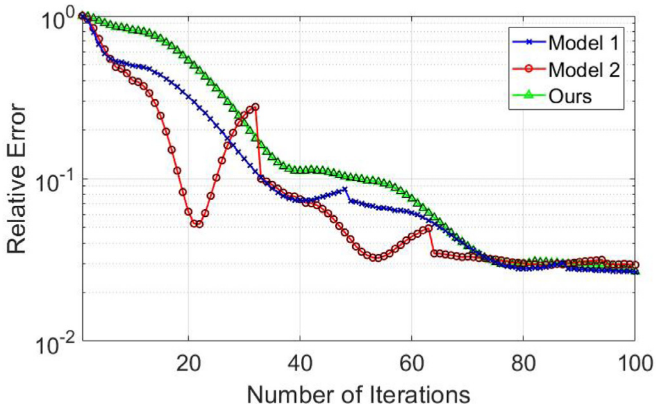


Fig. 5. The relative error of the deformation model in each iteration step. The descent method with hyperelastic material model is used in model 1. The descent method with anisotropy and viscoelasticity is used in model 2. In our deformation model, the constraints are incorporated. The relative error is defined as $(\epsilon(\boldsymbol{x}^{(l)}) - \epsilon(\boldsymbol{x}^{*}))/(\epsilon(\boldsymbol{x}^{(0)}) - \epsilon(\boldsymbol{x}^{(*)}))$, where $\boldsymbol{x}^{(l)}$ is the result in the lth iteration and $\boldsymbol{x}^{*}$ is the ground truth calculated by Vega [36].

The solution of our developed deformation model can converge to the exact result of the implicit Euler integration method given sufficient number of iterations. In order to verify the accuracy of the developed model, the relative error in each iteration step is calculated. The relative errors of our developed deformation and the descent method without considering constraint are shown in Fig. 5. Simulation results show that the solution of our developed can converge as fast as the descent method without considering constraint. In our developed method, 96 iterations are performed to get the final solution of the vertex position in each time step.

4.4. Simulation of neurosurgery

The deformation model proposed in this paper is incorporated into the development of a neurosurgical simulator. Similar to [13], the FEM-based model with optimization implicit method is used to simulate the deformation of brain tissue. Anisotropic and viscoelastic properties of the brain tissue are incorporated into the deformation model. In this paper, constraints of distance and permanent deformation are incorporated into the deformation model. Neurosurgical procedures such as microdissection and hemostasis are simulated, which can help young neurosurgeons to gain surgical experience through practising with the neurosurgical simulator. The Mimics, which is a software for image processing, three-dimensional designing and modeling, is used to reconstruct the geometric models of brain and meningioma through CT images of patients. The three-dimensional geometric model is converted into tetrahedral elements with TetGen, which is a mesh generator developed by Si [37]. As the tumor adheres to the brain tissue, the pairs of attached vertices in the geometric models are selected manually.

As shown in Fig. 6, the simulation of microdissection of a meningioma is presented. There are 2179 nodes and 7461 tetrahedral elements in the brain tissue model, and 1773 nodes and 5708 tetrahedral elements in the meningioma model. The convexity meningioma is attached on the surface of the brain tissue. Distance constraints are incorporated into 475 pairs of attached vertices, and the initial distance d_i is set to 1 mm. In the beginning of meningioma dissection, the tumor is retracted by a gasper to expose the tumor-brain interface. The scalpel is used to separate the tumor from the brain tissue through cutting the arachnoid bands within the tumor-brain interface. Distance constraints that imposed on the attached vertex pairs are removed step by step, as shown in Fig. 6(b) and (c). Finally, the tumor model is separated from the brain tissue model.

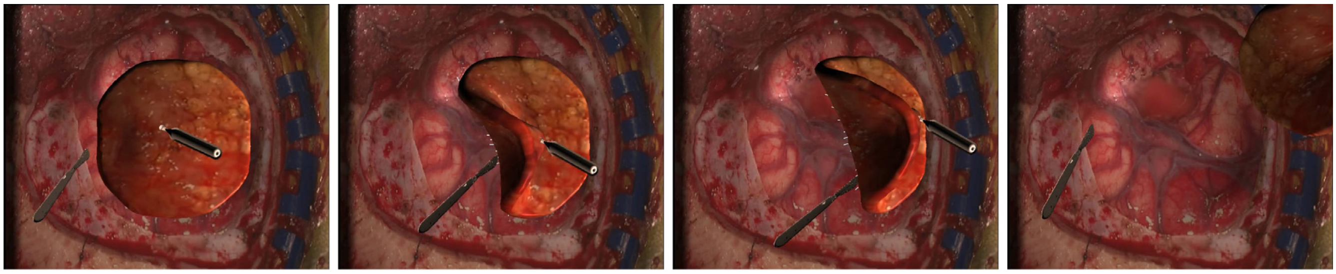
After the meningioma is removed, bipolar is usually used to perform hemostasis. Simulation of the hemostasis procedure is shown in Fig. 7. The bleeding site is identified firstly. Then the bipolar forceps is gently applied over the bleeding site and the heat generated by the bipolar forceps results to cauterization of bleed vessel, which leads to permanent deformation of brain tissue because of high temperature. As shown in Fig. 7(b), a virtual surgical instrument is used to cauterize the bleed vessel as well as the brain tissue. Collision detection between the virtual surgical instrument and the brain tissue is performed. The contact point is identified when the virtual surgical instrument touches the brain tissue. The permanent deformation is imposed on the brain tissue model. The simulation results of hemostasis is shown in Fig. 7(d).

Several simulators for neurosurgery training have be proposed [17,38]. NeuroTouch is a physics-based virtual simulator which provides a number of training tasks such as tumor-debulking, tumor cauterization, endoscopic nasal navigation and microdissection. Compared with those simulators using FEM model and implicit integration scheme, the neurosurgical simulator developed using our proposed deformation model achieves a higher computational efficiency. The neurosurgical simulator we developed is shown in Fig. 8. We integrated the simulated surgical scenes with real surgical ones, making it more realistic. The 3–D scene of the brain tissue can be observed on a 3-D monitor inside the cabinet. Two Phantom Omni desktop interfaces are used as the haptic control consoles and a CHAI3D open source software package is employed to drive the haptic interface.

5. Discussion

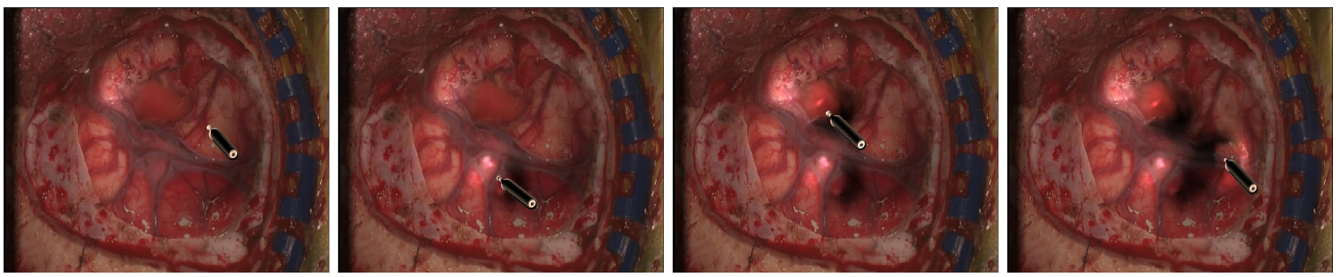
The simulation of constraints with an energy function in the proposed model is analogous to the penalty method [39]. As shown in equation (9), penalty forces are incorporated into the dynamic system of soft tissue. The penalty forces compete with all other forces applied on the vertices, which makes the dynamic system to be conformed by the constraints. When the constraints are violated, the penalty forces will pull the vertex back to its desired state.

However, the conventional penalty method [39] suffers the instability problem when the restoring force is large. On the other hand, the restoring force needs to be large enough in order to overpower all other competing forces. This instability problem is well avoided in our proposed model since the implicit integration scheme used is unconditionally stable. Moreover, the proposed



(a) (b) (c) (d)

Fig. 6. Simulation of neurosurgical procedures base on augmented reality: dissection of convexity meningioma.



(a) (b) (c) (d)

Fig. 7. Simulation of neurosurgical procedures base on augmented reality: hemostasis.

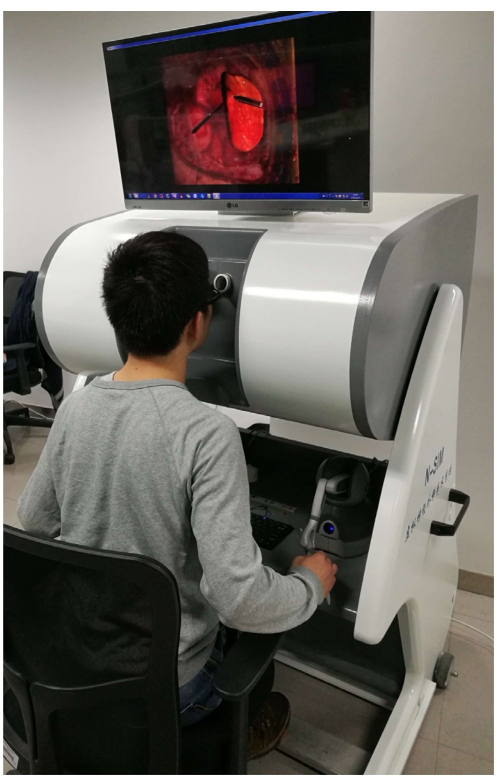


Fig. 8. The virtual-reality based neurosurgical simulator we developed.

model is relatively easy to implement and computationally efficient, making it suitable for simulating neurosurgical procedures such as the dissection of convexity meningioma and hemostasis.

The simulation of constraints with an energy function, however, may have the same drawback as the penalty method does [21]. Note that the parameter γ represents the stiffness of constraints and the restoring force applied on each vertex contains one or more stiffness parameters. It is difficult to tune stiffness parameters in a complex system with many competing forces. The penalty forces introduced into the dynamic system can also lead to undesired oscillations. To solve this problem, the method of Lagrange multipliers [40] can be used to calculate these parameters explicitly.

6. Conclusion

This paper presents a new brain tissue deformation model with constraints for neurosurgical simulation. The deformation model is based on the FEM model using the optimization implicit Euler method. An energy function of constraint that characterizes the interactions between the virtual instrument and soft tissue is incorporated into the deformation model. Distance and permanent deformation constraints are proposed to characterize the interactions in the convexity meningioma dissection and hemostasis. The proposed deformation model has been incorporated into the development of a neurosurgical simulator. The simulation results show that the proposed deformation model provides a high degree of visual realism and computational efficiency. While the proposed method is focused on the simulation of neurosurgical procedures, it can be well extended to other surgical procedures, such as laparoscopic surgery.

There are still a few types of interactions in neurosurgical simulation that are not explicitly modeled in the proposed model, such as ventriculostomy and tumor debulking. This is one direction that we will look at in the future work.

Conflict of interest

None.

Acknowledgments

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cmpb.2019.03.018.

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